

Kuiyi (Cory) Gao

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Mandarin (native) | English — TOEFL iBT 110 | Cantonese (intermediate)

Education

University of North Carolina at Chapel Hill

Jan 2026 – present

Undergraduate; intended B.A. in Computer Science

Expected May 2027

- GPA: 3.68/4.00. Exchange student in Spring 2026; transferred in August 2026.
- Coursework: COMP 550 Algorithms and Analysis, COMP 562 Machine Learning, COMP 586 Natural Language Processing, COMP 380 Technology, Ethics & Culture.

The Chinese University of Hong Kong

Sep 2023 – May 2026

Undergraduate study in Computer Science; transferred to UNC

- Faculty of Engineering ELITE Stream. Coursework: ELEG 5491 Introduction to Deep Learning (graduate level), ESTR 3108 Artificial Intelligence, Data Structures, Formal Languages & Automata Theory, Computer Organization & Design, Probability & Statistics, Linear Algebra.

Publications and Manuscripts

[1] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. ACL 2025 (Main Conference). [\[paper\]](#) [\[code\]](#)

[2] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. Findings of ACL 2025. [\[paper\]](#) [\[code\]](#)

[3] Co-first-authored manuscript on *GUI-agent robustness under interface noise*. Under review, 2026.

Research Experience

GUI-Agent Robustness under Interface Noise [3]

Hong Kong & remote

CUHK; advisor: Prof. Michael R. Lyu

Sep 2025 – present

- **Co-first author.** Co-developed a benchmark for web and Android tasks with interface changes such as overlays, moving content, and altered layouts; compared task completion, unsafe actions, and action sequences.
- Designed metrics for success within a retry limit, relative performance loss, unsafe actions, and changes in action sequences; helped implement paired tests on original and altered interfaces.

Runtime Safety for Third-Party Agent Skills [\[code\]](#)

Abu Dhabi & Beijing

MBZUAI UGRIP and continuing research; advisor: Dr. Steve Liu

Jun 2026 – present

- Built the runtime monitor in a four-student UGRIP team. It compared declared skill behavior with observed activity and produced PASS, DEFER, or BLOCK decisions with supporting logs.
- Developing typed Tickets: structured records linking independently approved permissions, observed activity, and decisions so auditors can recompute verdicts. Manuscript in preparation.

Research Experience (continued)

Chain-of-Jailbreak: Image Generation Safety [2]

Tencent AI Lab & CUHK

Hong Kong

Jun 2024 – Feb 2025

- **Second author.** Designed the method, contributed benchmark data, and ran most of the safety and defense experiments on multi-turn image editing.
- Built evaluation pipelines for model responses and a prompting-based defense; contributed to the manuscript and reviewer responses.

MMSafeAware: Multimodal Safety Awareness [1]

Tencent AI Lab

Hong Kong & Shenzhen

Dec 2023 – Oct 2024

- **Third author.** Contributed evaluation design and benchmark construction to a study of 1,500 image-prompt pairs across 29 safety scenarios, covering unsafe and benign inputs.
- Ran vision-centric reasoning fine-tuning experiments and evaluated missed unsafe inputs separately from false alarms on benign inputs. The overall study covered nine multimodal models and three mitigation approaches.

Selected Projects

MemSlot: Saliency for Long-Document Compression [code]

2026

UNC COMP 586, Natural Language Processing

- Implemented memory-slot attention over frozen RoBERTa features in PyTorch, using reconstruction and slot-diversity losses; merged scores from overlapping token windows into word-level saliency.
- Built document rendering guided by saliency scores for CUAD contracts, with OCR-decoded text sent to a separate QA reader and text-pruning and summarization baselines. End-to-end evaluation is not yet complete.

Poetry and Image Variation

2025

CUHK ENGE3940, AI and Language

- Analyzed saved CLIP-based scores of differences among images generated from ten Imagist and Romantic poetry excerpts, with 20 images per excerpt from each of FLUX.1-dev and Stable Diffusion.
- Used Python to plot score distributions, compare poem groups, and examine the relationship between prompt length and image variation.

Text-KEN: Kernel Comparisons for Sentence Embeddings

May – Aug 2025

CUHK summer research; advisor: Prof. Farzan Farnia

- Built a Python prototype to compare two sets of MiniLM sentence embeddings using RBF kernels and spectral entropy; explored centering and covariance whitening.

Technical Skills

Programming and data Python, NumPy, pandas, SciPy; Java and C (coursework).

ML and NLP PyTorch, Hugging Face Transformers, sentence-transformers; attention modules, frozen encoders, and fine-tuning experiments.

Evaluation Benchmark construction; task completion and action-sequence metrics; embedding comparisons; span-QA exact match/F1.

Visualization Matplotlib, seaborn, Pillow; statistical plots, document rendering, and image grids.

Research tools Docker, Playwright, Git, HTTP/JSON APIs; experiment configuration and trace logging.

Honors and Awards

Professor Joseph Lau Scholarship of Innovation; CUHK College Entrance Scholarship; CUHK exchange scholarship (UNC, Spring 2026).